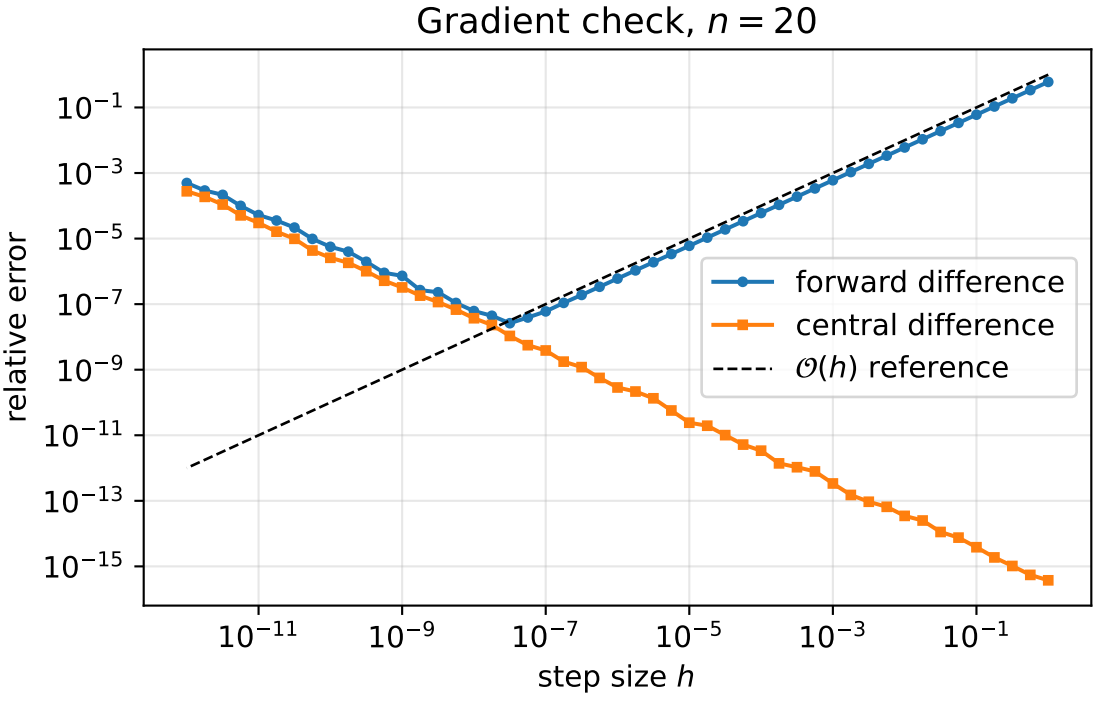
Gradient check, $n = 20$
relative error
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
10^{-11}
10^{-9}
10^{-7}
10^{-5}
10^{-3}
10^{-1}
step size h
forward difference
central difference
$\mathcal{O}(h)$ reference